Spatio-Temporal Aware Knowledge Graph Embedding for Recommender Systems

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***Abstract*—Knowledge Graphs (KGs) have been incorporated into recommender systems as side information to solve the classical data-sparsity and cold-start problems, with explanations for recommended items. Traditional embedding-based recommender systems generally utilize abundant information from KGs directly to enrich the representation of items or users, but the influences of spatial and temporal dependencies are usually ignored among them. In this paper, we propose a Spatio-Temporal Aware Knowledge Graph Embedding (STAKGE) for recommender systems, which incorporates spatio-temporal information with bias when propagating potential preferences of users in knowledge graph embedding. Moreover, we construct a multi-source KG-based recommender dataset – YelpST, containing spatio-temporal information. The experiments on YelpST dataset show that our proposed approach can capture comprehensive spatio-temporal correlations and improve the prediction performance as compared to various state-of-the-art baselines.**

***Index Terms*—POI, Recommender System, Knowledge Graph**

I. Introduction

With the growth of online content and services, recommender systems (RS) aim at recommending one or a set of unobserved items to users. To satisfy user potential requirements, collaborative filtering (CF) algorithm [1], [2] in RS predicts the most relevant items according to user-item historical interactions like purchases and clicks into filtering process. However, CF-based methods often suffer from data-sparsity [3] and cold-start problems [4], leading to the inclusion of side information such as user reviews, feedback, and item attributes in recent researches.

In particular, knowledge graphs (KGs) have been utilized as side information for recommender systems with respect to their structured knowledge, since they represent facts in structured forms with entities as nodes and relations as edges [5]. Therefore, KG-based recommender systems map users, items and their attributes, as well as relations between users and items, which can not only cope with data-sparsity problem, but also make the prediction of user preferences more accurate and explainable [6], [7].

However, some KG-based recommender systems only directly use abundant information from KGs to enrich the representation of items or users, and ignore other potential influences in KG for recommendation, such as space-time disciplines. For example, table 1 shows four facts related with *user1* in Spatio-Temporal Aware Knowledge Graph. *User1* choosed the nearby restaurant *Caffe Demetre* many times, while the restaurant *Twisted Meltz* near *Caffe Demetre* were also choosed in the later time.

In this paper, to capture hierarchical interests and spatio-temporal information automatically in KGs, we propose Spatio-Temporal Aware Knowledge Graph Embedding (STAKGE) for Recommender Systems. The key idea of STAKGE is to aggregate and incorporate spatio-temporal information with bias when propagates user potential preferences in KGs. Remarkably, STAKGE can (1) automatically capture long-range interests through preference propagation, and (2) weight neighbors based on the spatio-temporal information of KGs.

Furthermore, some researchers propagate user potential preferences and explore their hierarchical interests for recommendations on two commonly used benchmark datasets (i.e., Amazon-Book [8] and MovieLens-1M [9]). However, both of the benchmark datasets lack spatial information, which is important for practical applications. Instead, we construct a new KG-based recommender dataset YelpST by intersecting Yelp and Yelp2018 (Wang et al. [10]). Yelp is the largest review site in the United States, and Yelp2018 is adapted from the 2018 edition of the Yelp challenge. Wherein, local businesses such as restaurants and bars are viewed as items, which form a network of local business information derived as a knowledge graph. Yelp2018 contains 42 relations, 136499 entities as well as 1853704 triples, and we further construct

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TABLE I
FOUR FACTS RELATED WITH USER 1 IN SPATIO-TEMPORAL AWARE KNOWLEDGE GRAPH

User	Name	Location		Date
		Latitude	Longitude	
user1	Caffe Demetre	43.720040	-79.456428	2014/08/25 20:34:34
user1	Caffe Demetre	43.720040	-79.456428	2012/12/16 15:19:26
user1	Twisted Meitz	41.153530	-81.356878	2014/08/28 22:43:59
user1	Twisted Meitz	41.153530	-81.356878	2014/09/26 18:11:21

YelpST by intersecting Yelp2018 and Yelp to add spatial and temporal information. YelpST contains 42 relations, 136499 entities as well as 1853704 triples, and we use it as a benchmark in this paper. The experimental results show that STAKGE achieves 93.5% on AUC for recommendation, and outperforms state-of-the-art baselines.

The rest of this paper is organized as follows: Section 2 introduces works related to KG-based methods for recommendation. Section 3 presents the problem formulation. Section 4 describes our recommendation model STAKGE in detail. Section 5 discusses our experiments and results. Finally, we conclude the future work in Section 6.

II. RELATED WORK

KG-based recommendation methods can usually be classified into three categories: embedding-based methods [11]–[13], path-based methods [14]–[16], and the unified methods [6], [17].

Embedding-based recommendation methods obtain low-dimensional feature embeddings of *entities* and *relations* by the *knowledge graph embedding* (KGE) [18], [19] on a KG, and uniformly input embeddings into recommendation algorithms [20]. Deep Knowledge-aware Network (DKN) [12] models news by combining textual embedding of sentences learned with Kim CNN [21] and knowledge-level embedding of entities in news content via TransD [22]. Collaborative Knowledge base Embedding (CKE) [13] combines a CF module with knowledge embedding, text embedding, and image embedding of items in a unified Bayesian framework. Signed Heterogeneous Information Network Embedding (SHINE) [11] builds a sentiment network for users and targets, and then utilizes their social network and profile network as side information. Therefore, these are embed into deep autoencoders for recommendation. Embedding-based recommendation methods build KGs with multiple types of item information to enrich the representation of items, and such information can be used to model user representations more precisely. However, those methods mainly focus on modeling rigorous semantic relatedness (e.g., TransE [23] and TransR [24] assume *head* + *relation* = *tail*), which are more suitable for in-graph applications such as KG completion and link prediction than for recommendation.

Path-based recommendation methods build a user-item graph and leverage connectivity patterns of entities in the graph for recommendation [25]. Knowledge-aware Path Recurrent Network (KPRN) [16] encodes the extracted path sequence with an LSTM [26] layer and user preferences for items, and then the score in each path is aggregated via a weighted pooling layer for recommendation. Personalized Entity Recommendation (PER) [14] and Meta-Graph Based Recommendation [15] treat KG as a heterogeneous information network (HIN), and extract latent features from meta-path/meta-graph to represent the connectivity between users and items along different types of relation paths/graphs. Path-based methods utilize the structural information of KG in a more effective and intuitive way. However, these methods rely heavily on manually extracted features to represent path semantics, so they can not be applied to the scenarios (e.g., news recommendation) where entities and relations belong to different domain.

The unified methods are combinations of embedding-based methods and path-based methods. To fully exploit the information in KGs for better recommendations, the unified methods automatically discover potential user interests by integrating semantic representations of entities and relations as well as the connectivity information. For example, RippleNet [6] and AKUPM [17] exploit KG structure by propagating embeddings on it, which conduct to express user potential preferences in the KG. However, note that the spatio-temporal information in KG is also weakly characterized in them, while human behavior in real life conforms to certain spatio-temporal disciplines.

III. PROBLEM FORMULATION

We formulate KG-based recommender problem as follows. In a typical recommender scenario, we have a set of M users $\mathcal{U} = u_1, u_2, ..., u_M$ and a set of N items $\mathcal{V} = v_1, v_2, ...V_N$. The user-item interaction matrix $Y \in \mathbb{R}^{M \times N}$ is defined according to user implicit feedbacks, where $y_{uv} = 1$ indicates that user u engages with item v, such as clicking, browsing, or purchasing; otherwise $y_{uv} = 0$. Additionally, we also have a knowledge graph $\mathcal{G}$, which is comprised of entity-relation-entity triples (h, r, t). Here $h \in \mathcal{E}$, $r \in \mathcal{R}$, and $t \in \mathcal{E}$ denote the head, relation, and tail of a knowledge triple, $\mathcal{E}$ and $\mathcal{R}$ are two sets of entity and relation in the knowledge graph, respectively. For example, the triple (*Waffle house*, *Open24Hours*, *Jack in the Box*) states the fact that both *Waffle house* and *Jack in the Box* will be open 24 hours a day.

Given interaction matrix Y as well as knowledge graph $\mathcal{G}$, we aim to predict whether user u has potential interest in item v with which he has had no interaction before. Our goal is to learn a prediction function $\hat{y}_{uv} = \mathcal{F}(u, v; \Theta)$, where $\hat{y}_{uv}$ denotes the probability that user u will be interested in item v, and Θ denotes the model parameters of function $\mathcal{F}$. Note

TABLE II
SYMBOL IDENTIFICATION AND DESCRIPTION

Symbol	Description	Symbol	Description
$\mathcal{U}$	the set of users	$\mathcal{V}_u$	the historical set of interests for user u
$\mathcal{V}$	the set of items	$\mathcal{E}u^k$	the set of k-hop relevant entities for user u
Y	the user-item interaction matrix	$\mathcal{S}_u^k$	the set of knowledge triples that are k-hop(s) away from the seed set $\mathcal{V}_u$
$\mathcal{G}$	knowledge graph	$\hat{y}_{uv}$	the probability that user u will click item v
$\mathcal{E}$	the set of entities	$\mathcal{F}(u, v; \Theta)$	the prediction function
$\mathcal{R}$	the set of relations	Θ	the model parameters of function $\mathcal{F}$

that the important mathematical symbols with descriptions in this paper are shown in Table 2.

IV. STAKGE

In this section, we propose STAKGE model to capture spatio-temporal information for recommendation based on preference progagation. Furthermore, we present a learning algorithm for spatio-temporal information. At last, we provide some discussions on the complete learning algorithm of STAKGE.

A. Framework

As shown in Figure 1, STAKGE takes a user u and an item v as input, and outputs the predicted probability that user u will click item v. For the input user u, the set of historical interest $\mathcal{V}_u$ is treated as seeds in KG, then extended along links to form multiple ripple sets $\mathcal{S}_u^k (k = 1, 2, ..., H)$. A ripple set $\mathcal{S}_u^k$ is the set of knowledge triples that are *k*-hop(s) away from the seed set $\mathcal{V}_u$.

Specifically, each item v is associated with an item embedding $\mathbf{v} \in \mathbb{R}^d$, where d is the dimension of embeddings. Given the item embedding $\mathbf{v}$ and the 1-hop set of knowledge triples $\mathcal{S}_u^1$, each triple (h_i, r_i, t_i) in $\mathcal{S}_u^1$ is assigned a relevance probability by comparing item v to the head h_i and the relation r_i in this triple:

$$p_i = softmax(\mathbf{v}^{\mathrm{T}} \mathbf{R_i} \mathbf{h_i}) = \frac{exp(\mathbf{v}^{\mathrm{T}} \mathbf{R_i} \mathbf{h_i})}{\sum_{(h,r,t) \in \mathcal{S}_u^1} exp(\mathbf{v}^{\mathrm{T}} \mathbf{R} \mathbf{h})} \quad (1)$$

where $\mathbf{R_i} \in \mathbb{R}^{d \times d}$ and $\mathbf{h_i} \in \mathbb{R}^d$ are the embedding of relation r_i and the embedding of head entity h_i, respectively. The relevance probability p_i can be regarded as the similarity of item embedding $\mathbf{v}$ and the entity embedding $\mathbf{h_i}$ measured in the space of relation $\mathbf{R_i}$. Note that it is necessary to take the embedding matrix $\mathbf{R_i}$ into consideration when calculating the relevance of item embedding $\mathbf{v}$ and entity embedding $\mathbf{h_i}$, since an item-entity pair may have different similarities when measured by different relations. For example, "*Forrest Gump*" and "*Cast Away*" are highly similar when considering their directors or stars, but have less in common if measured by genre or writer.

After obtaining the relevance probabilities, we take the sum of tails in $\mathcal{S}_u^1$ weighted by the corresponding relevance probabilities, and the vector o_u^1 is returned:

$$\mathbf{o}_u^1 = \sum\nolimits_{(h_i, r_i, t_i) \in \mathcal{S}_u^1} p_i \mathbf{t}_i \quad (2)$$

where $\mathbf{t_i} \in \mathbb{R}^d$ is the embedding of tail t_i. Vector $\mathbf{o}_u^1$ can be seen as the 1-order response of user u click history $\mathcal{V}_u$ with respect to item v, This is similar to item-based CF methods, in which a user is represented by his related items rather than a independent feature vector to reduce the size of parameters. Through Eq. (1) and Eq. (2), a user's interests are transferred from his history set $\mathcal{V}_u$ to the set of his 1-hop relevant entities $\mathcal{E}_u^1$ along the links in $\mathcal{S}_u^1$, which is called preference propagation in STAKGE.

Note that by replacing $\mathbf{v}$ with $\mathbf{o}_u^1$ in Eq. (1), we can repeat the procedure of preference propagation to obtain the 2-order response $\mathbf{o}_u^2$ of user u, and the procedure can be performed iteratively on ripple sets $\mathcal{S}_u^i$ for $i = 1, 2, ..., H$ of user u. Therefore, a user's preference is propagated up to H hops away from his historical preference, and we observe multiple responses of user u with different orders: $\mathbf{o}_u^1$, $\mathbf{o}_u^2$, ..., $\mathbf{o}_u^H$. The embedding of user u with respect to item v is calculated by combining the responses of all orders:

$$\mathbf{u} = \sum_{i=1}^{H} \mathbf{o}_u^i \quad (3)$$

Though the user response of last hop $\mathbf{o}_u^H$ contains all the information from previous hops theoretically, it is still necessary to incorporate $\mathbf{o}_u^k$ of small hops k in calculating user embedding since they may be diluted in $\mathbf{o}_u^H$. Finally, the user embedding and item embedding are combined to output the predicted clicking probability:

$$\hat{y}_{uv} = sigmoid(\mathbf{u}^T \mathbf{v}) \quad (4)$$

B. Spatio-temporal Information Extraction

The spatio-temporal disciplines of human activities play an important role in recommendation process. For example, in business recommendation, people usually choose nearby restaurants, and the restaurants people once choose may also be chosen several times in the future. Therefore, this paper analyzes human activities, and excavates the geographical aggregation effect as well as temporal effect. The dataset of this paper is from Yelp, a merchant review website. There are 45919 users with 1177186 reviews about 136499 businesses

1) Spatial Effect: Numerous studies have demonstrated that human behavior obeys power-law distribution, and this spatial pattern is also known as *Levy-flight* pattern because it is similar to animal foraging behavior [28]. In this paper, the distance from the user location to the destination is taken as travel distance, and the relation between user access intention and

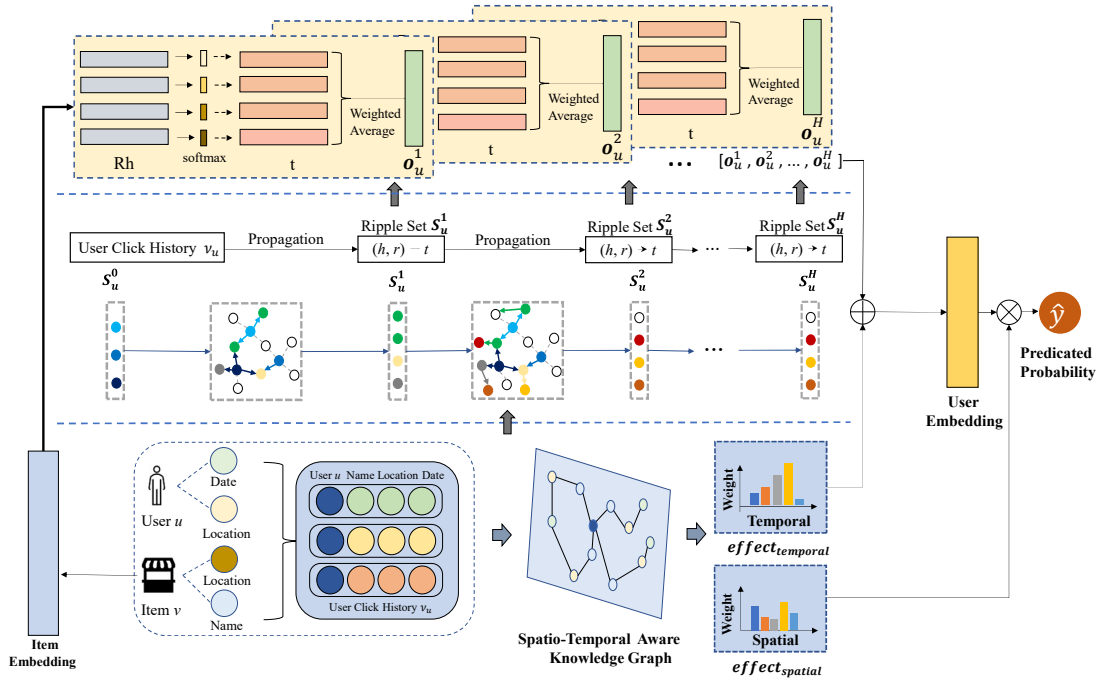


Fig. 1. The framework of STAKGE.

the travel distance is approximated by a power-law distribution model. Therefore, in the output layer of access probability calculation, we consider power-law attenuation caused by travel distance, and enhance Eq. (4) to obtain Eq. (5):

$$\hat{y}_{uv} = sigmoid(\mathbf{u}^T\mathbf{v} \cdot effect_{spatial}) \tag{5}$$

$$effect_{spatial} = a_1 \cdot dist^{b_1} + c_1 \tag{6}$$

where $effect_{spatial}$ is the power-law attenuation term of travel distance $dist$, and a_1, b_1, c_1 are the parameters of attenuation term.

2) Temporal Effect: We consider temporal effect in STAKGE since user interests may change over time. Therefore, we use logarithmic decay function to model the migration of user preferences. For the temporal effect, we add the set of $\mathcal{V}_u$ after logarithmic decay into the combining the of all orders, and enhance Eq. (3) to obtain Eq. (7):

$$\mathbf{u} = \mathbf{o}_{\boldsymbol{u}}^{\mathbf{0}} \cdot effect_{temporal} + \sum_{i=1}^{H} \mathbf{o}_u^i \tag{7}$$

$$effect_{temporal} = e^{-a_2(t-t_0)} \tag{8}$$

where $effect_{temporal}$ is the logarithmic attenuation term of engaging time t of user u, t_0 is the current time, and a_2 is the parameters of the attenuation term.

V. Experiments

In this section, we demonstrate the performance of our proposed STAKGE on item recommendation, and discuss the effectiveness of our design choices along with key factors. We first introduce the datasets, baselines, and experiment setup, then present the experiment results. We will also give a case study of visualization and discuss the choice of hyper-parameters in this section.

A. YelpST Dataset

Recent approaches (Wang et al. [6]; Wang et al. [10]) for KG-based recommendation often use Amazon-Book and MovieLens-1M as benchmarking datasets. We advocate that those datasets can not fully capture the task complexity, since they lack spatial information which is also important for recommendations.

To alleviate the above problem, we propose a new dataset, YelpST, extracted from Yelp2018. The following steps are used to sample our dataset from Yelp2018. We begin with a set of seed nodes corresponding to entities from Yelp2018 having a direct mapping in **Yelp** (*business_id*). Then, we connect the *latitude* and *longitude* values of **Yelp: business** to the statements whose $main$ object correspond to **Yelp: business**. Similarly, connect the $date$ values to the Yelp2018 dataset. This step results adding spatial and temporal information on Yelp2018 dataset.

To evaluate the effectiveness of STAKGE, we conduct experiments on one benchmark dataset: YelpST, which is publicly accessible and vary in terms of domain, size, and sparsity. The statistics are summarized in Table 3 and the details are introduced as follows.

Since YelpST is explicit feedback data, we transform it into implicit feedback where each entry is marked with 1 indicating that the user has rated the item (the threshold of rating is 3 for YelpST), and sample an unwatched set marked as 0 for each

TABLE III
BASIC STATISTICS OF THE THREE DATASETS.

Dataset	# Users	# Items	# Interactions	# Relations	# KG triples	Size (MB)
Yelp	1637138	192609	6685900	—	—	8845.45
Yelp2018	45919	136499	1177186	42	1853704	143.55
YelpST	45919	136499	1177186	42	1853704	37.38

user, which is of equal size with the rated ones. For YelpST, we use the ID embeddings of users and items as raw input.

For each dataset, we randomly select 80% of interactions of each user to constitute the training set, 10% of interactions as validation set to tune hyper-parameters, and treat the remaining as the test set.

B. Baselines and experimental settings

We compare the proposed STAKGE with the following baselines, in which the first two baselines are KG-free while the rest are all KG-based methods. Hyper-parameter settings for baselines are introduced in the next subsection.

- **SVD** [24] is a classic CF-based model using inner product to model user-item interactions.
- **LibFM** [3] is a feature-based factorization model in CTR scenarios. We concatenate user ID and item ID as the input for LibFM.
- **LibFM + TransE** extends LibFM by attaching an entity representation learned by TransE [23] to each user-item pair.
- **PER** [27] treats KG as heterogeneous information networks and extracts meta-path based features to represent the connectivity between users and items.
- **CKE** [14] combines CF with structural, textual, and visual knowledge in a unified framework for recommendation. We implement CKE as CF plus a structural knowledge module in this paper.
- **RippleNet** [6] is a memory-network-like approach that propagates users' pre-ferences on the KG for recommendation.

For all baselines and our method STAKGE, we set the hop number $H = 2$ for YelpST. A larger number of hops hardly improves performance, but does incur heavier computational overhead according to the experiment results. The complete hyper-parameter settings are given in Table 4, where d denotes the dimension of embedding for items and the knowledge graph, and η denotes the learning rate. The hyper-parameters are determined by optimizing AUC on a validation set. For fair consideration, the latent dimensions of all compared baselines are set the same as in Table 4, while other hyper-parameters of baselines are set based on grid search.

We implement our STAKGE in Python. The algorithms are executed on an operating system of Ubuntu 20.04, with GeForce RTX 3080 and 32 GB RAM. Each experiment is repeated 3 times, and the average performance is reported. We evaluate our method in two experiment scenarios: (1) In click-through rate (CTR) prediction, we apply the trained model to each piece of interactions in the test set and output the predicted click probability. We use Accuracy and AUC to evaluate the performance of CTR prediction. (2) In top-K recommendation, we use the trained model to select K items with highest predicted click probability for each user in the test set, and choose $Precision@K$, $Recall@K$, $F1@K$ to evaluate the recommended sets.

TABLE IV
HYPER-PARAMTERS SETTINGS FOR THE YELPST DATASET.

Para	Description	Setting
d	Dimension of embeddings	8
H	Depth of receptive field	3
K	Neighbor sampling size	4
λ	L2 regularizer weight	10^{-7}
η	Learning rate	2×10^{-2}

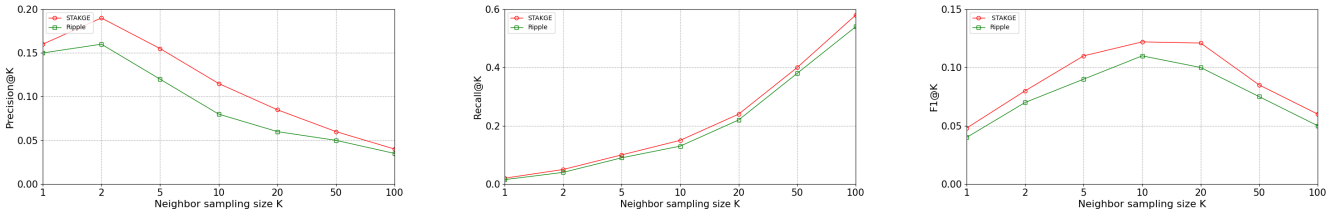
TABLE V
THE RESULTS OF AUC AND ACCURACY IN CTR PREDICTION.

Model	AUC	Accuracy
SVD	86.1	79.8
LibFM	86.4	81.0
LibFM + TransE	87.3	82.1
PER	82.8	75.1
CKE	80.9	78.9
RippleNet	89.9	84.9
STAKGE	**93.5**	**87.3**
SAKGE	92.2	86.2
TAKGE	91.6	85.7

The results of CTR prediction and top-K recommendation are presented in Table 5 and Figure 2, respectively. We have the following observations:

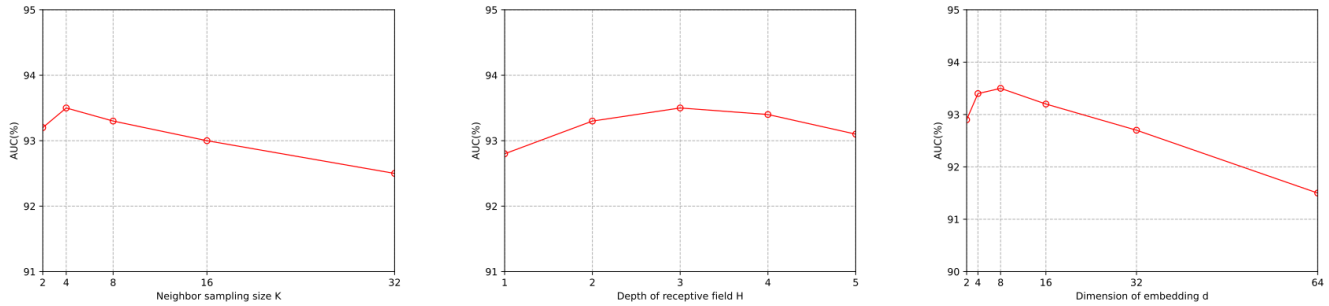
- In general, STAKGE achieves the best performance on the YelpST dataset, outperform the second best method by 2.4% on Accuracy and 3.6% on AUC. STAKGE also achieves outstanding performance in top-K recommendation as shown in Figure 2.
- Compared with KG-free models SVD and LibFM, the main improvement of STAKGE comes from automatically propagating user potential preferences and exploring their hierarchical interests in KG.
- Compared with KG-based models (LibFM + TransE, PER, CKE, RippleNet), the reason that STAKGE is superior is twofold. First, STAKGE excavates the relation between user access intention and the travel distance. Second, it also integrates the temporal effect, such the current demands are explicitly guided by historical habits.

The last three rows in Table 4 summarize the performance of STAKGE variants. The SAKGE removes the temporal dimension from STAKGE, treating YelpST as static KG, while the next variant TAKGE removes the spatial dimension. From



(a) (b) (c)

Fig. 2. The results of top-K recommendation for YelpST



(a) (b) (c)

Fig. 3. AUC result of STAKGE with different hyper-parameters

the results we find that STAKGE outperforms both variants on AUC and Accuracy, so we make the following observations:

- Compared with SAKGE, the modeling of temporal preferences can improve the performance, which also validates the assumption that the immediate decision of users is guided by their historical habits.
- Compared with TAKGE, the improvements in STAKGE demonstrate the effectiveness of spatial embedding.

1) Impact of neighbor sampling size: We vary the size of sampled neighbor K to investigate the efficacy of usage of the KG. From Figure 3(a) we observe that STAKGE achieves the best performance when $K = 4$. This is because a too small k does not have enough capacity to incorporate neighborhood information, while a too large K is prone to be misled by noises.

2) Impact of depth of receptive field: We investigate the influence of depth of receptive field in STAKGE by varying H from 1 to 4. The results are shown in Figure 3(b). We observe the occurrence of serious model collapse when $H = 3$, as a larger H brings massive noises to the model. This is also in accordance with our intuition, since a too long relation chain makes little sense when inferring inter-item similarities. An H of 2 is enough for real cases according to the experiment results.

3) Impact of dimension of embedding: Lastly, we examine the influences of embedding dimension d on performance of STAKGE. The result in Figure 3(c) is rather intuitive: increasing d initially can boost the performance since a larger d can encode more information of users and entities, while a too large d adversely suffers from overfitting.

VI. Conclusion and Future Work

This paper proposes STAKGE for recommender systems. STAKGE overcomes the limitations of existing KG-based methods by propagating user potential preferences selectively and biasedly, which is able to excavate the spatio-temporal effects. The extensive experimental results on YelpST dataset demonstrate the recommendation effectiveness of STAKGE.

For future work, we plan to (1) focus on modeling user characteristics; (2) design an algorithm to well capture the impact of weekdays and weekends on the recommendation process.

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